Reader: A limit is a number to which a given sequence tends (converges).

Author: What do you mean by saying "converges to a number"?

Reader: I mean that with an increase of the serial number, the terms of a sequence converge very closely to a certain value.

Author: What do you mean by saying "very closely"?

Reader: Well, the difference between the values of the terms and the given number will become infinitely small. Do you think any additional explanation is needed?

Author: The definition of the limit of sequence which you have suggested can at best be classified as a subjective impression. We have already discussed a similar situation in the previous dialogue.

Let us see what is hidden behind the statement made above. For this purpose, let us look at a rigorous definition of the limit of sequence which we are going to examine in detail.

Definition: The number a is said to be the limit of sequence (y_n) if for any positive number ε there is a real number N such that for all $n > N$ the following inequality holds: $|y_n - a| < \varepsilon$ (1)

↳ V. Imp Definition

Reader: I am afraid, it is beyond me to remember such a definition.

Author: Don't hasten to remember. Try to comprehend this definition, to realize it's structure and its inner logic. You will see that every word in this phrase carries a definite and necessary content, and that no other definition of the limit of sequence could be more succint (more delicate even).

First of all, let us note the logic of the sentence. A certain number is the limit provided that for any $\varepsilon > 0$ there is a number N such that for all $n > N$ inequality (1) holds. In short, it is necessary that for any ε a certain number N should exist.